

for chronic limb-threatening ischemia (CLTI). We aim to compare outcomes of bypasses for CLTI based on preoperative beta-blocker use.

Methods: We used Vascular Quality Initiative data for lower extremity bypasses performed for CLTI (2011-2023). Patients were stratified by preoperative beta-blocker use. The primary outcome was 30-day mortality. Secondary outcomes included postoperative complications, return to the operating room (RTOR), prolonged length of stay (PLOS; ≥ 7 days), postoperative major amputation, and major adverse cardiovascular event (MACE). MACE was defined as myocardial infarction, stroke or in-hospital death. A secondary analysis of one-year outcomes was performed in patients on chronic beta-blocker use stratified by discharge beta-blocker continuation. The χ^2 test, Kaplan-Meier survival estimates, and logistic and Cox regressions were used for analyses.

Results: A total of 37,703 limbs from 30,560 patients were studied: chronic preoperative beta-blocker use (for >30 days) (20,231, 53.7%), no beta-blocker use (15,305, 40.6%), beta-blocker use only on operative day (803, 2.1%), and beta-blocker use for 1 to 30 preoperative days (1,364, 3.6%). Patients in the >30 day cohort were more likely to be older and have severe comorbidities. After adjusting for confounders, the pattern of preoperative beta-blocker use was not associated with 30-day mortality. However, no beta-blocker use was associated with decreased risk of stroke, MACE, and renal complications compared to >30 -days (Table).

Additionally, beta-blocker use only on operative day and 1 to 30 days preoperatively were associated with increased risk of PLOS and RTOR, while 1 to 30 day use was also associated with an increased risk of renal complications (Table). In the 1-year analysis, patients with chronic preoperative beta-blocker use that discontinued beta-blockers after discharge had increased hazards of 1-year mortality and combined mortality/major amputation compared to continued use after discharge (Fig).

Conclusions: We found that beta-blocker use for >30 days was associated with poor outcomes compared to no beta-blockers due to higher age and comorbidities. Interestingly, we found that starting beta-blockers within 1 month of surgery was associated with a higher risk of RTOR, PLOS, and renal complications. Additionally, continuing beta-blocker use following discharge in patients who were on chronic use before surgery was associated with improved overall and amputation-free survivals. We conclude that in patients without chronic beta-blocker use, initiation of beta-blockers before bypass is not beneficial. However, patients with chronic preoperative beta-blocker use benefit from continuation of beta-blockers after bypass. Further prospective studies are necessary to confirm our findings.

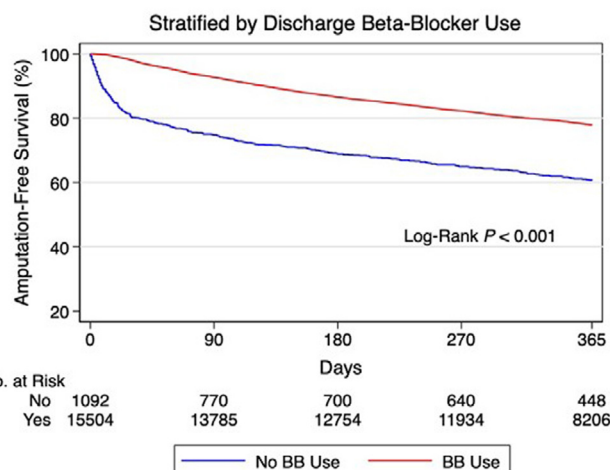


Fig. One-year amputation-free survival of patients with chronic preoperative beta-blocker use (>30 -day) following lower extremity bypass stratified by beta-blocker use following discharge.

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PC226

Comparison of Bovine Carotid Artery Grafts versus ePTFE Grafts for Dialysis Access in a Rural Population

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Objectives: Native arteriovenous fistulas are the preferred method for dialysis access but are not feasible for all patients. This study aims to compare the outcomes of bovine carotid artery (BCA) grafts and expanded polytetrafluoroethylene (ePTFE) grafts in a predominantly rural dialysis population.

Table. Multivariate analyses of perioperative and one-year outcomes of patients with no preoperative beta-blocker use versus chronic beta-blocker use (>30 days) as well as operative-day only use and 1- to 30-day use versus no beta-blocker use

Outcome	No BB vs BB for >30 days Reference = >30 days		Op-day BB vs No BB Reference = No BB		BB for 1-30 days vs No BB Reference = No BB	
	aOR (95% CI)	P value	aOR (95% CI)	P value	aOR (95% CI)	P value
30-Day mortality	0.97 (0.83-1.14)	.730	1.26 (0.76-1.11)	.374	1.13 (0.76-1.68)	.374
In-hospital death	0.82 (0.67-1.01)	.0061	0.92 (0.42-2.02)	.841	1.33 (0.86-2.05)	.206
MI	0.89 (0.76-1.04)	.134	1.08 (0.57-2.04)	.805	1.16 (0.81-1.66)	.429
Stroke	0.76 (0.59-0.98)	0.032	0.21 (0.03-1.47)	.116	1.31 (0.67-2.57)	.429
MACE	0.87 (0.77-0.99)	0.041	0.97 (0.57-1.66)	.915	1.16 (0.88-1.52)	.293
Respiratory complications	0.85 (0.71-1.00)	.0050	1.28 (0.76-2.15)	.356	1.37 (0.93-2.03)	.112
Renal complications	0.75 (0.68-0.83)	<.001	1.38 (0.95-1.99)	.088	1.45 (1.15-1.82)	.001
RTOR	1.01 (0.94-1.08)	.799	1.41 (1.16-1.72)	.001	1.48 (1.23-1.79)	<.001
PLOS	0.99 (0.94-1.05)	.827	1.22 (1.01-1.49)	.043	1.56 (1.34-1.82)	<.001
Postoperative MA	1.00 (0.83-1.20)	.970	1.10 (0.65-1.86)	.731	1.05 (0.65-1.71)	.836

aOR, Adjusted odds ratio; BB, beta-blocker; CI, confidence interval; MA, major amputation; MACE, major adverse cardiovascular event; MI, myocardial infarction; Op-day, operative day; PLOS, prolonged length of stay; RTOR, return to operating room. Boldface entries indicate statistical significance.

Methods: Patients undergoing hemodialysis access creation with either BCA or ePTFE grafts from 2013 to 2019 were identified in our Integrated Health Care System Vascular Quality Initiative database. Kaplan-Meier analysis was used to evaluate primary, primary assisted, and secondary patency. Additional outcomes, including the number of interventions, rates of infection, pseudoaneurysm, arterial steal syndrome, graft abandonment, and mortality, were also assessed.

Results: A total of 188 patients were analyzed: 94 received BCA grafts and 94 received ePTFE grafts. No statistically significant differences were observed in primary, primary assisted, or secondary patency between groups, although secondary patency tended to favor the BCA group. Patients with ePTFE grafts required significantly more interventions per graft to maintain patency compared to the BCA group (2.6 vs 1.5; $P = .006$). Graft abandonment was more common in the ePTFE group (60.6% vs 39.4%; $P = .005$). While infections were more frequent in the ePTFE group (7.45% vs 2.13%), the difference did not reach statistical significance. Rates of pseudoaneurysm, arterial steal syndrome, and death were similar between the groups.

Conclusions: In our predominantly rural dialysis population, BCA grafts demonstrated comparable patency outcomes to ePTFE grafts with fewer required interventions and a lower rate of graft abandonment. These findings suggest that BCA grafts may offer clinical advantages over ePTFE grafts. Future cost analyses are warranted to evaluate the potential economic benefits of BCA graft use.

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PC090



Covered vs Uncovered Stents for Aortoiliac Occlusive Disease in the Vascular Quality Initiative

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Objectives: There remain conflicting data regarding the advantages of covered stents (CS) vs uncovered stents (US) for aortoiliac occlusive disease. The COBEST trial reported higher patency rates of CS particularly in TransAtlantic Inter-Society Consensus class C/D lesions, whereas subsequent studies such as the DISCOVER trial did not find significant differences. The Gore VBX FORWARD study is currently enrolling to compare these two modalities. As such, we sought to compare the outcomes between US and CS in the Vascular Quality Initiative (VQI).

Methods: All patients undergoing US or CS with or without plain balloon angioplasty of the common iliac, external iliac, or both arteries for aortoiliac occlusive disease between 2016 and 2023 were identified in the VQI. Patients with prior inflow stenting, prior ipsilateral surgical bypass, or concomitant procedures such as an atherectomy were excluded. One-to-one propensity score matching was performed using patient comorbidities, demographics, medications, and lesion characteristics as covariates. The primary outcome included primary patency at 1-year, defined as no evidence of $>50\%$ stenosis, occlusion, and without a target lesion revascularization. Secondary outcomes included 5-year mortality and technical success. Outcomes were evaluated using χ^2 tests, Kaplan-Meier estimates, and log-rank tests.

Results: Overall, 6881 patients underwent US and 2245 underwent CS. After matching, 1722 pairs of patients were included in our study with no baseline differences between the two groups (US: mean [SD] age, 66.1 [10.0] years; 56% male; CS: 66.1 [10.4] years; 57% male). Target lesion location was similar: common iliac (76% US vs 76% CS), external iliac (16% vs 15%), and both common and external (8% vs 9%; $P = .6$) (Table). Indication was also similar: claudication (71% vs 71%), rest pain (18% vs 18%), and tissue loss (11% vs 11%; $P = .9$). Overall, there were no significant differences in primary patency at 1-year (92% vs 92%; $P = .8$; Fig) and 5-year mortality (93% vs 91%; $P = .4$). For TransAtlantic Inter-Society Consensus class C/D lesions, primary patency at 1-year remained similar (92% vs 91%; $P = .5$). No differences in primary patency were seen for patients treated for claudication (94% vs 94%; $P = .9$) vs rest pain/tissue loss (89% vs 89%; $P = .8$). Technical success was the same between the groups (99% vs 99%; $P = .9$).

Conclusions: In this one-to-one propensity-matched retrospective analysis, US and CS had similar rates of primary patency, mortality, and technical success. Further long-term patency analysis is needed to elucidate the appropriate use of US and CS.

Table. Procedural characteristics after 1:1 propensity score matching

	US ^a (n = 1722)	CS ^{aw} (n = 1722)	P value ^b
Artery location			.56
Both common and external iliac	133 (7.7%)	150 (8.7%)	
Common iliac	1316 (76.4%)	1306 (75.8%)	
External iliac	273 (15.9%)	266 (15.4%)	
Preoperative symptoms			.85
Claudication	1222 (71.0%)	1226 (71.2%)	
Rest pain	304 (17.7%)	310 (18.0%)	
Tissue loss/ulcer/necrosis	196 (11.4%)	186 (10.8%)	
Urgency			.92
Elective	1600 (92.9%)	1598 (92.8%)	
Emergent	11 (0.6%)	13 (0.8%)	
Urgent	111 (6.4%)	110 (6.4%)	
Missing	0 (0%)	1 (0.1%)	
TransAtlantic Inter-Society Consensus Class			.76
A	724 (42.0%)	736 (42.7%)	
B	520 (30.2%)	501 (29.1%)	
C	295 (17.1%)	287 (16.7%)	
D	183 (10.6%)	198 (11.5%)	

CS, Covered stent; US, uncovered stent.

^aData are presented as number (%).

^b χ^2 test.

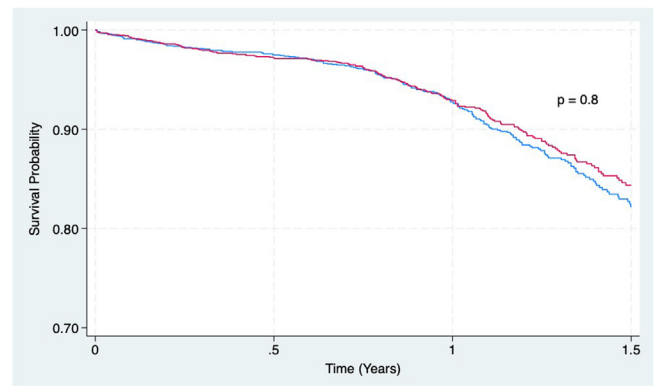


Fig. Primary patency by stent type.

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Salvage of Infected Aortobifemoral Bypass With Bilateral Obturator Bypasses

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